

SAKHAWAT HOSSAN ROBEL

sakhawathossanrobels@gmail.com | [Google Scholar](#) | [LinkedIn](#) | Sylhet, Bangladesh

PROFESSIONAL SUMMARY

A mechanical engineer with strong experience in computer simulation, focusing on mechanical design and modeling, computational fluid dynamics, molecular dynamics, and finite element analysis. Hands-on experience in applying modeling and simulation techniques in mechanics of nanomaterials, polymer electrolytes, electrochemistry, biomaterials, fluid-structure interaction, and thermal management systems, such as phase change materials and passive cooling systems. Gained experience in research methodologies, literature review, scientific writing, technical presentation, and problem-solving through multiple research assistant roles during my undergrad. Driven by curiosity, I am looking for an experimental and advanced simulation-driven mechanical/materials Ph.D. opportunity to become an independent researcher and contribute to science.

EDUCATION

B.Sc. in Mechanical Engineering

March 2022 – June 2026

Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh

- Senior year thesis: Computational Investigation of Dopant-Induced Mechanical, Thermal, and Fracture Characteristics of Ti-, Si-, and Ti-Si-Doped Monolayer Graphene via Molecular Dynamics Simulation.
- Junior year project: Development and fabrication of an Arduino-based autonomous fire-fighting robot capable of independently detecting and extinguishing fire without human intervention.

RESEARCH EXPERIENCES

Undergraduate Research Assistant

January 2023 – June 2024

Colorado State University, Fort Collins, Colorado, USA

Advisor: Dr. Nipa Khair, Lead Biomaterials Engineer, [YoungHeartValve Inc.](#)

- Developed CAD models of fiber-reinforced polymeric heart valve in SolidWorks to upgrade valve durability and hemodynamic performances.
- Analyzed SEM image of hyaluronan-based biomaterial particle size using ImageJ to validate synthesis efficiency.
- Contributed to data visualization and data analysis.

Undergraduate Research Assistant

November 2024 – May 2025

Shahjalal University of Science and Technology, Sylhet, Bangladesh

Advisor: Dr. Nuruzzaman Sakib, Associate Professor, Mechanical Engineering

- Developed computational models for N, B, B-N, Si doped graphene material using LAMMPS scripts to investigate mechanical strength, stiffness, and thermal transport as a function of dopant concentration.
- Performed Molecular Dynamics-based fracture analysis to characterize failure mechanisms in functionalized graphene, contributing to nanomaterial design for next-generation thermal and structural applications.

Head, Mechanical Design Team

January 2024 – September 2025

SUST Auto Drive – An Autonomous Vehicle Project of Shahjalal University of Science and Technology

- Led the mechanical design team consists of 5 sophomore students, in CAD modelling with GD&T principles, and fabrication of the chassis of an autonomous passenger vehicle.
- Collaborated with the sensor design and integration team to ensure design for manufacturing.

Research Assistant

May 2025 - Present

Shahjalal University of Science and Technology, Sylhet, Bangladesh

Advisor: Mohammad Junaid, Lecturer, Mechanical Engineering

- Developing a 3D CFD model in ANSYS Fluent to evaluate dual-layer PCM roofs for passive cooling in hot-humid climates.
- Simulate coupled heat transfer, convection, and solid-liquid phase change using real climatic data from major Bangladeshi cities like Dhaka, Cox's Bazar, and Jashore.

PUBLICATIONS

Peer-reviewed journal article

- [1] Md Junaid, **Sakhawat Hossan Robel**, Mahmud Or-Rashid, Pabel Shahrear, Md Shajedul Karim, Suvash C Saha, Goutam Saha "CFD Analysis of Dual-Layer PCM Roofs for Thermal Mitigation in Residential Buildings." *Heat Transfer, Wiley*, Jan 2026. DOI: <https://doi.org/10.1002/htj.70191>
- [2] Nipa Khair, Sanchita Bhat, **Sakhawat Hossan Robel**, Srujana Joshi, Katie Vinterella, Lakshmi Dasi, Susan James "Parametric Finite Element Modeling of Reinforced Polymeric Leaflets for Improved Durability." *Journal of Mechanical Behavior of Biomedical Materials, Elsevier*, Jan 2025. DOI: <https://doi.org/10.1016/j.jmbbm.2024.106884>
- [3] **Sakhawat Hossan Robel**, Mohammad Junaid, Pabel Shahrear, Md Shajedul Karim, Suvash C. Saha "Thermal Performance of PCM-Integrated Roofs for Passive Cooling in Tropical Climates: A 3D Numerical Study" [Submitted in "Building Simulation" – a Springer Journal]
- [4] **Sakhawat Hossan Robel**, Kamrul Hasan. "Enhancing Ionic Transport in Cross-linked PEO–LiTFSI Polymer Electrolytes: A Molecular Dynamics Study" [In progress]

Conference article

- [1] **Sakhawat Hossan Robel**, Nuruzzaman Sakib. "Effect of Si Doping on Mechanical, Thermal, and Fracture Behavior of Monolayer Graphene: A Molecular Dynamics Approach." *15th International Conference on Mechanical Engineering, BUET*, Dec 2025. DOI: <https://doi.org/10.2139/ssrn.6201278>
- [2] Nipa Khair, Sanchita Bhat, **Sakhawat Hossan Robel**, Srujana Joshi, Katie Vinterella, Lakshmi Dasi, Susan James. "TCT-875 Parametric Finite Element Modeling of Reinforced Polymeric Leaflets for Improved Durability." *Journal of the American College of Cardiology*, Oct 2025. DOI: <https://www.jacc.org/doi/abs/10.1016/j.jacc.2025.09.1062>

PROFESSIONAL TRAINING AND CERTIFICATION

- Industrial Training — Shahjalal Fertilizer Company Limited (SFCL): Hands-on training in large-scale chemical process control, plant operations, safety protocols, and quality systems in an industrial setting.
- CSWA: Certified SolidWorks Associate — Mechanical Design

- 3Dswymer: 3DEXPERIENCE Associate — Collaborative CAD platform

LEADERSHIP & SERVICE

- Assistant General Secretary (8th Executive Committee, 2024–2025), Binary SUST — an officially recognized student-led technology organization (est. 2015) at SUST, fostering computational literacy through workshops, coding contests, and training programs in programming, data science, machine learning & AI.
- Executive Member (7th Executive Committee, 2023–2024), contributing to workshop planning and student technical outreach initiatives.
- Delivered instructional sessions on systematic literature review and research methodology selection for junior students — organized by the Mechanical Engineering Association of SUST.

EXPERTISE & SKILL

Domain	Tool/Language	Methodological competencies
Modeling & Simulation	ANSYS Fluent	Conjugate heat transfer, solid-liquid phase change, turbulence modeling (k- ϵ , k- ω SST), transient thermal boundary conditions
	LAMMPS	MD force field scripting (AIREBO, Tersoff), tensile loading, and thermal conductivity calculation
CAD	SolidWorks, AutoCAD	2D drawing, 3D modeling and assembly, GD&T, design for manufacturing (DFM)
Programming	Python	Data processing and visualization, materials modeling
	MATLAB	ODE solving, signal, and data visualization
	Arduino	Automation & control
Data visualization and analysis	OVITO	MD trajectory analysis, defect identification, atomic strain visualization, and fracture analysis
	Tecplot	CFD and MD post-processing
	ImageJ	Microscopy image analysis, particle sizing, microstructure quantification